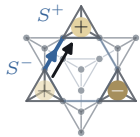


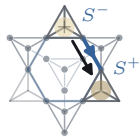
i.a) create


 $S_i^+ S_j^-$: pair splits

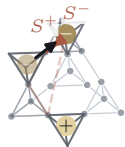
i.b) move


 $S_i^+ S_j^-$: spinon walks

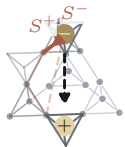
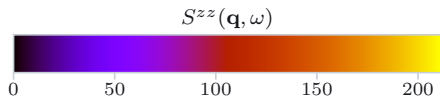
i.c) annihilate


 $+$ meets $-$
 $\mathbf{q}_\gamma = (0, 0, 0)$

i.d) create


 $S_i^+ S_j^-$: pair splits
i.e) apply $S_i^+ S_j^-$ 
charge on t_2 moves past t_5 , PBC maps it back onto t_4
out of the drawn cell

i.f) identify


 $\mathbf{q}_\gamma = (0, 0, -1)$


iii.a) periodic ED



iii.b) winding-free

